Deep learning techniques for classification of brain MRI

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Abstract: Several brain diseases are becoming a threat to the livelihood of people. One such problem is the presence of a brain tumour. A brain tumour can be benign or malignant. It is dangerous if it is a malignant or secondary tumour (metastasis). Therefore, there is a need to detect the presence of tumours at the earliest stage as possible. Using an automated method for brain tumour detection can be a solution to medical expertise as a biopsy can be excluded if early detection could be possible. Classification helps in the prediction of the type of image and type of tumour. In this paper, three stages are involved. In the first stage, the classification of brain MR images into normal (tumour) or abnormal (non-tumour) images using ConvNet, Lenet, ResNet, and Densenet has been analysed. In the second stage, architectures like Lenet and Alexnet are used in the prediction of the type of tumour

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namely metastasis, glioma, and meningioma. And lastly, using U-Net and AlexNet, classification into high grade glioma and low grade glioma was done.

Keywords: convolutional neural network; CNN; DenseNet; ResNet; AlexNet; U-Net.

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1 Introduction

Mitosis is a process of making new body cells (https://www.britannica.com/science/mitosis). When mitosis is not regulated correctly, health problems such as cancer may result (https://www.britannica.com/science/mitosis). These abnormalities need to be detected at an early stage so that prevention and cure could be provided. Researchers have found that most of the cells that are developed abnormally occurs in the brain (Pandian and Kumari, 2016). Therefore there is a need to detect abnormalities in the brain for accurate and effective decision making by a practitioner. A person suspecting of having any abnormalities is asked to undergo certain tests. There are plenty of imaging modalities tests that are used for scanning of the brain. These are magnetic resonance imaging (MRI), single photon emission computed tomography (SPECT), computed tomography (CT), positron emission tomography (PET), etc. (Schapira et al., 2017). Here, MRI images are considered due to their power and ability to create detailed images. It allows the physician to see any part of the body in a narrow slice. It is a non-invasive test, i.e., radiation free. It is also the safest mode of imaging modality being used (Mayfield, 2018).

For abnormalities detection, images are read one by one (slice by slice) thoroughly for obtaining images that contain abnormalities. This process of reading images and decision making consume the effective time of physicians. This issue can be solved by the use of automated techniques. Automated techniques have evolved in the past years starting with simple image processing techniques to statistical analysis which eventually leads to a more advanced type of methods like machine learning (ML) (Burje et al., 2017; Wahlang et al., 2019; Yang et al., 2018; Manikandan and Devi, 2014). ML mimics the human way of thinking and thus makes a decision. It is a set of learning algorithms that can be divided into three categories

1 rule-based

2 artificial neural network (ANN)-based (deep learning)

3 Bayesian theorem (Lee et al., 2017).

These techniques are not only used for the detection of abnormalities but with a hope to attain efficient accuracy as well (Suhag and Saini, 2015).

In this paper, deep learning-based methodologies are considered due to their excellent performances in the field of image processing. Firstly, classification of MRI images into normal (tumorous) or abnormal (non-tumorous) classes were done. This is important to know whether the images contain abnormalities or not. Secondly,

classification of images into three types of tumour namely metastasis, glioma, and meningioma has been done. It is done as it is crucial to know the type of tumour. Next, classification of images into high grade glioma (HGG) and low grade glioma (LGG) was done. Grading of the tumour is essential as treatment can be provided accordingly. This is important because classification can be a base for segmentation as it acts as a source of prediction of the type of images. Once an image is known of having abnormalities, segmentation can be done, i.e., finding region or area of abnormalities (Pereira et al., 2016).

1.1 Biomedical data for brain MRI

Biomedical data can be in heterogeneous formats such as digital imaging and communications in medicine (DICOM), Nii (NIFTI1 data format), mnc file extension, etc. Many data are available online which can be used for a different purpose. Related to our work some of the datasets that can be used are found in many sites. As an example, brainweb data are classified into normal and lesion data which is in mnc format. BRATS data which is in NII format with its ground truth can be used for segmentation purposes and detection of the tumour. Others include SICAS Medical Image Repository, cancer imaging archive, figshare, etc. There is numerous site in which data can be obtained freely and can be used for research purposes.

1.2 Our contributions

- Classification using convolutional neural network (CNN)-based methodologies was done into three categories, which was not done before.
- Four architectures are used for classification into normal or abnormal, namely, ConvNet, LeNet, ResNet, and DenseNet.
- U-Net with SVM and AlexNet have been used for classification into HGG and LGG.
- AlexNet is used for classification into metastasis, glioma, and meningioma.

1.3 Organisation of the paper

In this paper, Section 2 includes works related to the classification and concept of CNN. Section 3 gives a brief discussion on classification of MRI using CNN-based models for all three categories, Section 4 provides the experimental result, and the conclusion of the paper is given in Section 5.

2 Related work

Classification-based literature presents a variety of techniques. Many of the popular classification techniques are supervised in nature with significant features obtained in advance using feature extraction techniques. These include techniques like support vector machine (SVM) (Balasubramanian and Sudha, 2014; Nelly et al., 2013; Taie and

Ghonaim, 2017; Zhang and Wu, 2012), neural network (NN) (Praveen and Agrawal, 2015; Rajesh and Malar, 2013; Al-Badarneh et al., 2012; Sharma et al., 2020; Guo et al., 2019), K-nearest neighbour (KNN) (Al-Badarneh et al., 2012), gradient-based kernel selection technique (Dogra et al., 2020), etc. A comparison of techniques using feature extraction techniques like GLCM (Taie and Ghonaim, 2017; Zhang and Wu, 2012) and DWT and GLCM hybrid can be seen in Wahlang et al. (2018). Most of the commonly used techniques are that of ML. Although these techniques have proved to be good in classification, accuracy, and efficiency could not be achieved yet. For classification into metastasis, glioma, and meningioma few existing techniques are there which include that of International MICCAI BraTS Challenge. 1-578 (2018) which is similar. Classification into HGG and LGG is done mainly in the BRATS challenge. Segmentation field is more explored than that of classification. Classification can help to reduce the effort for segmentation if an image can be predicted as having abnormalities at an early stage.

With the advance of ML, NN-based methodology like deep learning has outbreak other methodologies in medical imaging (Guo et al., 2019; Zhang et al., 2019; Meenakshi et al., 2017). Brain tumour classification using deep learning methodology into three categories has been done in this paper. Classification is a process in which objects are categorised into some classes (Praveen and Agrawal, 2015; Mareeswari et al., 2017). All the methods that have been used in this paper are CNN-based methodology. CNN is a special architecture of ANN. It is popularly used in image classification (Kumar Mallick et al., 2019; Noreen et al., 2020; Li et al., 2019). An image is passed through a set of convolution, pooling, and fully connected layer and then generate an output. The basic building blocks of CNNs, in general, is shown in Figure 1 and are described briefly as follows:

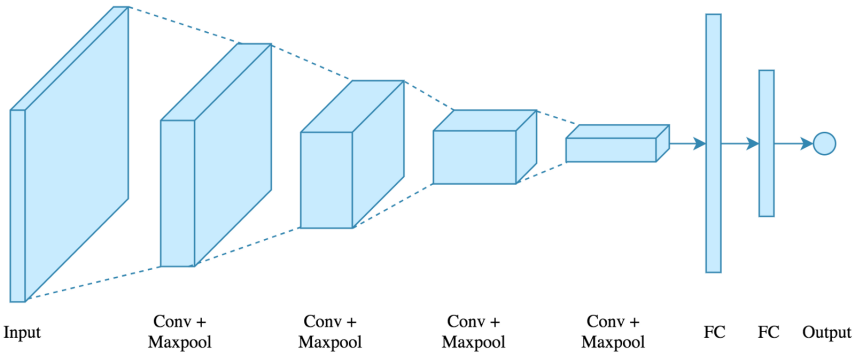
- Convolution layer: Convolution layer uses a set of convolution filters. This can be thought of as a NN with local connectivity and weight sharing with which the number of parameters is reduced. The role of a convolutional layer is to detect local features at each input maps. The convolution operation uses larger shifts and not like the traditional way where one position is shifted (Zhou et al., 2017). This is done with what commonly referred to as stride. Depending on the number of kernels or filters, the number of channels is counted. Depth relies on the number of channels. Depth can be 1 or 3 for greyscale and RGB images, respectively. The filter that is used is multiplied with the original image pixel which is summed up where only one pixel is obtained in the end.

- Pooling: Pooling layer role is to reduce the spatial size of feature maps (Zhou et al., 2017). It can be thought as dilation operation of image processing. Pooling can be max pooling, min pooling or average pooling. The most commonly used pooling is max pooling. Max pooling can be compared to max filter where the filter is replaced by its maximum value. Pooling layer performs a downsampling or subsampling operation along the spatial dimensions (width, height) (LeCun et al., 2015; Pan et al., 2015).

- ReLU transfer function: All deep networks usually consist of nonlinear function after convolution. Traditional NN uses tanh or sigmoid nonlinearity. In CNN rectified linear unit (ReLU) is used which is expressed as $ReLU(x)$ = max (0, x). This nonlinearity train faster than other. It makes the calculation much easier in back propagation because of its unity gradient (Mitchell, 1997).

- Fully connected layer: After convolution layer, pooling and nonlinearity (activation function), fully connected layer is followed. It is preferable if fully connected layers are used in lesser number as parameters will be increased with its usage (Zhou et al., 2017).
- Other important components are optimisation parameter, dropout and batch normalisation.

Figure 1 Convolutional neural network (see online version for colours)



Source: Dertat (2017)

3 Classification of brain MRI images using CNN-based model

Keeping in mind the importance of CNN, in our work four architectures namely ConvNet, Lenet, Resnet, and Densenet has been used for classification of brain images into normal or abnormal classes. AlexNet and LeNet for classification into metastasis, glioma, and meningioma and AlexNet and U-Net for classification into HGG and LGG were used. The architectures with their parameters are explained in brief next.

3.1 Classification into normal or abnormal brain images

In this section, architectures that are used in classification of brain MRI images into two classes, i.e., normal or abnormal are explained in brief.

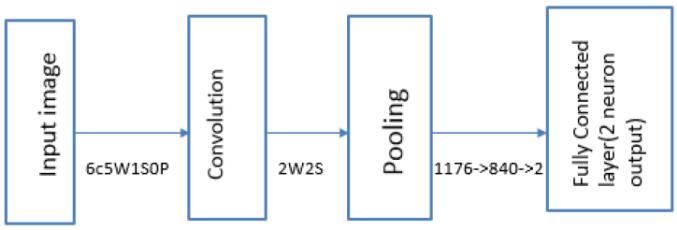
3.1.1 ConvNet

A simple CNN is used here which is ConvNet (LeCun et al., 2015). It consists of a convolutional layer and a pooling layer in the first stage. Features are learned in the first stage. A part of the image is connected to the next convolutional layer. The dot product is applied between the receptive field and the filter. The outcome of this is a single integer of the output volume. The output of the summation of all weights is passed through the ReLU activation function. The use of weight in CNN helps in

detecting patterns in different parts of the array. All these layers will be passed to a fully connected layer. The diagram is as shown in Figure 2.

In this paper the input image is of size 32 × 32, which gives a 6 channel output. Following this layer is the max pooling layer with a filter size of 2 × 2. Here, a stride of 2 is used. This is followed by ReLU nonlinearity and the output layer of 1,026 neurons is obtained in the fully connected layer. Two fully connected layers are applied. In the last fully connected layer two neurons are obtained one for each class, i.e., normal or abnormal.

Figure 2 ConvNet architecture (see online version for colours)

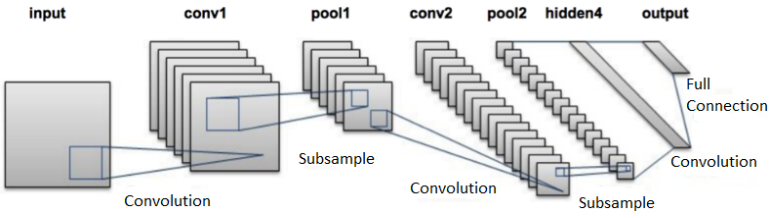


Source: LeCun et al. (2015)

3.1.2 LeNet

LeNet architecture was developed in LeCun et al. (2015) which can be seen in Figure 3 (Das, 2017). It was used by several banks for recognising handwritten numbers on the cheque. An input image of 32 × 32 greyscale image was used. A convolutional layer is applied with a kernel (filter) size of 5 × 5. In this stage, low level features (i.e., edges) were detected. It is followed by a max pooling layer with a stride of 2 for subsampling data. This is then followed by a second convolutional layer with a max pooling layer for obtaining higher level features. In every layer, the ReLU activation function is used. A dropout of 0.5 was taken to reduce overfitting. After pooling is applied, two fully connected layers are followed, wherein the last layer two neurons are obtained. The same structure is used in this paper.

Figure 3 LeNet architecture (see online version for colours)



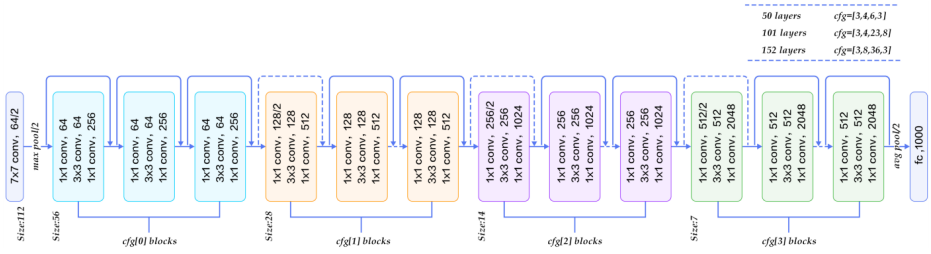
Source: Das (2017)

3.1.3 ResNet

Kaiming et al. (2016) proposed ResNet architecture. A novel architecture with 'skip connections' (Rajesh and Malar, 2013) and features heavy batch normalisation. ResNet solves the problem of vanishing input or gradient until it reaches the end of the network (Simonyan and Zisserman, 2014). ResNet uses residual connections unlike AlexNet and GoogleNet (Das, 2017).

ResNet takes an input size of 224 × 224 and convolved with 7 × 7 filter using a stride of 2. It is then followed by a max pooling operation. The output of the pooling layer is then passed to a residual block. ResNet makes it possible to train up to hundreds or even thousands of layers. In each residual block, three convolutional layers are there. In each convolutional layer, three consecutive convolution operation is applied with different filter size as could be seen from Figure 4. The dotted lines represent subsampling and solid lines of the residual connection. In this paper, ResNet of 152 layers was used. The architecture was used but not the trained weight. Here, we have modified into two class problem to suit our purpose.

Figure 4 ResNet architecture (see online version for colours)



Source: Das (2017) and Kaiming et al. (2016)

3.1.4 DenseNet

DenseNet architecture was proposed by Huang et al. (2018). Despite the tackling of vanishing gradient problem, this network also encourages feature reuse. It can differentiate between information that is preserved and added information. Each layer has direct access to the gradient from the loss function and the original image. This allows the training of deeper networks even further. It requires fewer parameters as redundant features are need not be learn. DenseNet combines features by the use of concatenation operation, unlike ResNet. Each layer obtains output and reads the state from preceding layers. The structure is similar to that of ResNet. Here a 1 × 1 convolutional layer is used before a 3 × 3 convolutional layer to reduce the number of feature maps. The overall architecture is shown in Table 1.

3.2 Classification into metastasis, glioma and meningioma

Classification is done into three types namely, metastasis, glioma, and meningioma. Metastasis is a secondary tumour. It is the most common lesion that occurs in the brain. It spreads from other parts of the body to the brain. The types most likely to cause brain

metastasis are lung, breast, colon, kidney, and melanoma (https://www.mayoclinic.org/diseases-conditions/brain-metastases/symptoms-causes/syc-20350136). Glioma is a type of brain tumour that grows from glial cells. Glioma is graded into four grades. Grade I is pilocytic astrocytoma, grade II is LGG, grade III is malignant glioma and grade IV is glioblastoma multiforme (https://www.mayfieldclinic.com/PE-Glioma.htm).

Table 1 DenseNet architecture

Layers	*Output size*	*DenseNet-121 (k = 32)*	*DenseNet-169 (k = 32)*	*DenseNet-201 (k = 32)*	*DenseNet-161 (k = 48)*
Convolution	112 ×112	7 × 7 conv, stride 2			
Pooling	56 × 56	3 × 3 max pool, stride 2			
Dense	56 × 56	1 × 1 conv × 6	1 × 1 conv × 6	1 × 1 conv × 6	1 × 1 conv × 6
block (1)		3 × 3 conv × 6	3 × 3 conv × 6	3 × 3 conv × 6	3 × 3 conv × 6
Transition	56 × 56	1 × 1 conv			
layer (1)	28 × 28	2 × 2 average pool, stride 2			
Dense	28 × 28	1 × 1 conv × 12	1 × 1 conv × 12	1 × 1 conv × 12	1 × 1 conv × 12
block (2)		3 × 3 conv × 12	3 × 3 conv × 12	3 × 3 conv × 12	3 × 3 conv × 12
Transition	28 × 28	1 × 1 conv			
layer (2)	14 × 14	2 × 2 average pool, stride 2			
Dense	14 × 14	1 × 1 conv × 24	1 × 1 conv × 32	1 × 1 conv × 48	1 × 1 conv × 36
block (3)		3 × 3 conv × 24	3 × 3 conv × 32	3 × 3 conv × 48	3 × 3 conv × 36
Transition	14 × 14	1 × 1 conv			
layer (3)	7 × 7	2 × 2 average pool, stride 2			
Dense	7 × 7	1 × 1 conv × 16	1 × 1 conv × 32	1 × 1 conv × 32	1 × 1 conv × 24
block (4)		3 × 3 conv × 16	1 × 1 conv × 32	3 × 3 conv × 32	3 × 3 conv × 24
Classification	1 × 1	7 × 7 global average pool			
layer		1,000D fully connected, softmax			

Source: Huang et al. (2018)

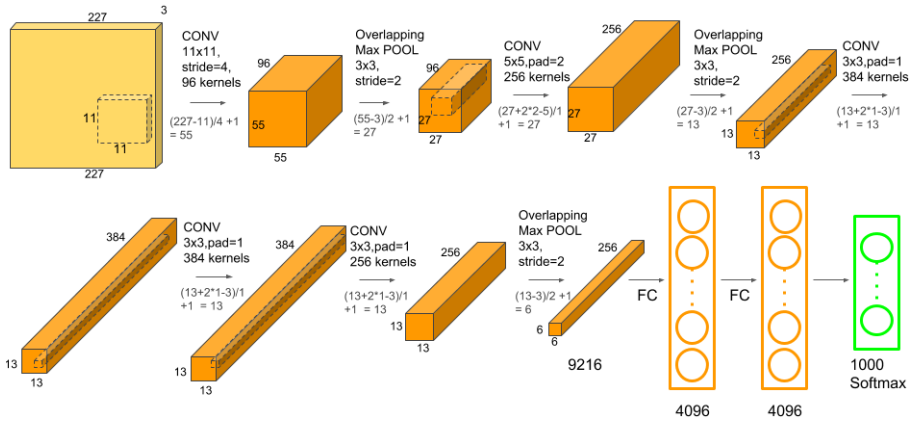
Meningioma is a type of tumour that is likely to be benign than malignant. It is a type of tumour that arises from the meninges (https://www.mayoclinic.org/diseases-conditions/meningioma/symptoms-causes/syc-20355643). It occurs mostly in women. It is important to know the type of tumour as it can be differentiated and proper diagnosis and prognosis could be provided at the right time. For classification into the three types of tumours, two methods have been analysed namely, AlexNet and LeNet.

3.2.1 Alexnet

AlexNet is designed by Alex Krizhevsky. It was the winning architecture of ImageNet in 2012. It is a CNN-based methodology, that solves the problem of image classification. It was originally used for the classification of cats and dogs. The input is an RGB image of size 256 × 256. The images were divided into train images and test images. AlexNet consists of five convolutional layers and three fully connected layers (https://www.learnopencv.com/understanding-alexnet/). In the first convolutional layer, there are 96 kernels of size 11 × 11 × 3, followed by the overlapping nax pooling layers. The third, fourth, and fifth convolutional layers are connected directly. ReLU nonlinearity is applied after all the convolutional and fully connected layers. ReLU is followed by normalisation before pooling. The overall architecture is as shown in Figure 5.

In this paper, the same architecture of AlexNet has been used with the same input size image. In the first layer 96 kernels of size 5 × 5 × 3 has been used, followed by normalisation and max pooling layer of size. Fully connected layers are changed based on the input of the previous layer. Data augmentation has been used in this case by means of cropping of images into different sizes. Input image of size 1,024 × 1,024 are cropped into 512 × 512 for experimental purposes. Max pool of 2 × 2 is used instead of 3 × 3. The number of kernels used is 96 and 256 only. The kernel of 384 has been kept as 256.

Figure 5 AlexNet architecture (see online version for colours)



Source: https://www.learnopencv.com/understanding-alexnet/

3.2.2 LeNet

In this case, the number of convolution layers is increased from 2 to 4. Here, the convolutional layer has a filter size of 5 × 5, followed by another convolution layer of the same kernel, for better feature learning. After this, the max pooling layer of size 2 × 2 and stride of 2 is used for subsampling. After which a third convolutional layer is applied and followed by a max pooling layer and ReLU activation function. The fourth convolution layer is the same as that of the third layer. The rest is the same as explained in the previous section.

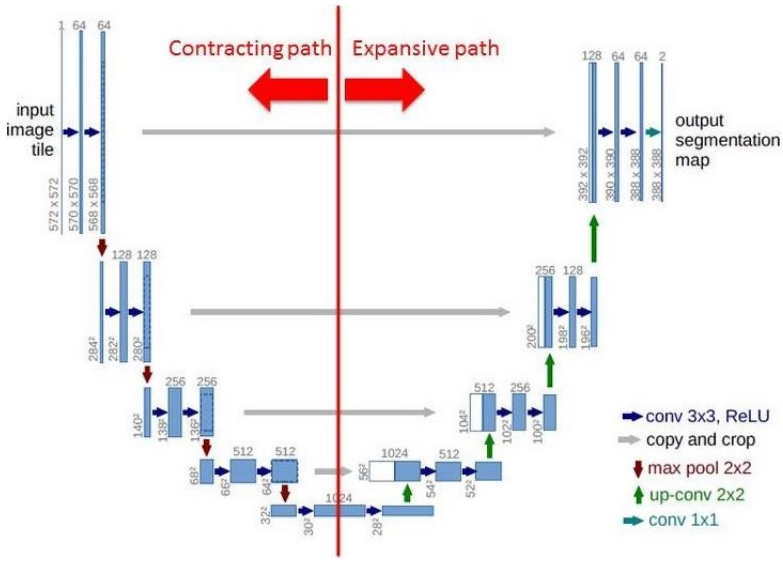
3.3 Classification into HGG and LGG

Classification into HGG and LGG is important in the treatment procedure of tumour. From an MRI scan, an image is predicted as having HGG or LGG. Glioma can be a threat to the survival rate of people. LGG can be treated and chances are high to get cured whereas HGG is more liable to be life threatening. Here, to solve this issue a convolutional net called U-Net is applied. It is taken because of its capability of giving good results in most cases. The architecture was modified for more precise segmentation even for few training images (https://en.wikipedia.org/wiki/U-Net). It is a method that is used mainly for segmentation purpose similar to SegNet. Unlike SegNet, U-Net works well for greyscale images. The architecture of U-Net is given below.

3.3.1 U-Net

Is a technique used for segmentation purpose. It is a fully convolutional network (FCN). It is symmetric as can be seen in Figure 6. Because of which it has a large number of feature maps in the upsampling path. It has three parts downsampling, bottleneck, and upsampling. Each block is composed of a convolutional layer and activation function (with batch normalisation) and max pooling. The number of features is doubled at each pooling layer. This happens so that image segmentation can be fulfilled. Bottleneck includes batch normalisation, convolutional layer, and dropout. Next, unsampling consists of deconvolution, convolutional layer, and activation function. It is used for precise localisation. In this type of net, parameters that are learned are kernel only (https://en.wikipedia.org/wiki/U-Net). The architecture of U-Net is shown in Figure 6.

Figure 6 U-Net architecture (see online version for colours)



Source: http://deeplearning.net/tutorial/unet.html

In this paper, U-Net architecture is used for greyscale images. Each block is composed of 3 × 3 convolution layer + activation function (with batch normalisation) and 2 × 2 max pooling (http://deeplearning.net/tutorial/unet.html). The advantage of U-Net is that image of any sizes can be used as an input. The same structure of U-Net has been used with ReLU nonlinearity. The output of U-Net is then fed to a linear SVM classifier for classifying into HGG or LGG.

4 Experimental results

The results are validated using parameters like accuracy, loss, and computation time (Suhag and Saini, 2015; Chowdhury et al., 2012) which are described below.

4.1 Accuracy

Accuracy is a metric used for evaluating classification models. It is the measure of prediction the model got right (Kashyap et al., 2017; https://developers.google.com/machine-learning/crash-course/classification/accuracy). Accuracy has the following definition:

$$Accuracy = \frac{Number\ of\ images\ correctly\ classified}{Total\ number\ of\ images} \tag{1}$$

4.2 Loss

Loss is a measurement of how well the model behaves after every iteration. The loss function used in our case is cross entropy with the following formula (https://ml-cheatsheet.readthedocs.io/en/latest/loss_functions.html):

$$-(y\log(p) + (1-y)\log(1-p)) \tag{2}$$

for two class classification and for three class classification it is given as

$$-\sum_{c=1}^{m} y_{o,c}\log(p_{o,c}) \tag{3}$$

where y is the indicator if c is correct classification for o and p is the predicted class.

4.3 Operation compute complexity

Compute complexity is the amount of resources needed for running an algorithm (https://en.wikipedia.org/wiki/Computational_complexity). It gives the measure of parameters learned in the whole algorithm (Huang et al., 2018). Mathematically it can be represented as:

$$Operation = 2cwh(M - w + 1)(N - h + 1) \tag{4}$$

where M and N is the size of the input image. w and h is the width and height of the filter, respectively, and c is the number of channel.

4.4 Classification into normal or abnormal

Two datasets are used for validation using data obtained from the SICAS Medical Image Repository and the other from the North Eastern Indira Gandhi Regional Institute of Health and Medical Sciences (NEIGRIHMS). From SICAS, 5,754 images are obtained, out of which 3,454 images are abnormal images (tumorous) and 2,300 are normal images (non-tumorous). From these, for training we have taken 3,000 abnormal and 2,000 normal images and for testing 454 abnormal and 300 normal images. The dataset obtained from the hospital constitutes 4,000 abnormal and 620 normal images. Out of which, 3,000 abnormal images and 500 normal images and 1,000 abnormal images and

120 normal images are taken for training and testing respectively. Dataset1 constitutes of testing data and dataset2 constitutes of training data.

It can be observed from Table 2 and Figure 7 that ResNet and DenseNet gives better output compared to other methods. These two Nets have a very deep structure and are expected to give better output (Huang et al., 2018). But due to insufficient data, the result is not as expected. Overall these two Nets show less loss and better accuracy. From the training loss parameter, we can say that the vanishing gradient problem has overcome.

Figure 7 Graph showing accuracy of training and testing for normal and abnormal classification (see online version for colours)

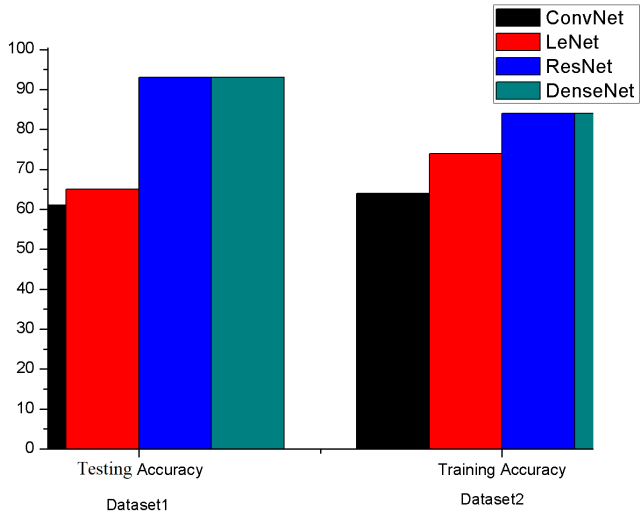


Table 2 Output of classification into normal or abnormal

CNN structure	*Dataset1*		*Dataset2*		*Computational complexity (approx)*
	Testing loss	*Testing accuracy*	*Training loss*	*Training accuracy*	
ConvNet	0.1	61%	0.1	64%	43 K
LeNet	0.1	65%	0.1	74%	86 K
ResNet	0.01	93%	0.01	84%	11 M
DenseNet	0.01	93%	0.01	84%	7 M

4.5 Classification into metastasis, glioma and meningioma

Using images obtained from Radiopaedia, classification into metastasis, glioma, and meningioma was done. The total number of patient data is 11 with each having 25–50 slices. Out of the total, 2, 6, and 3 patients are affected with metastasis, glioma, and meningioma, respectively. FLAIR MRI images are used for this purpose as it can give differentiation of abnormalities in a much better way. Prediction of type of tumour is

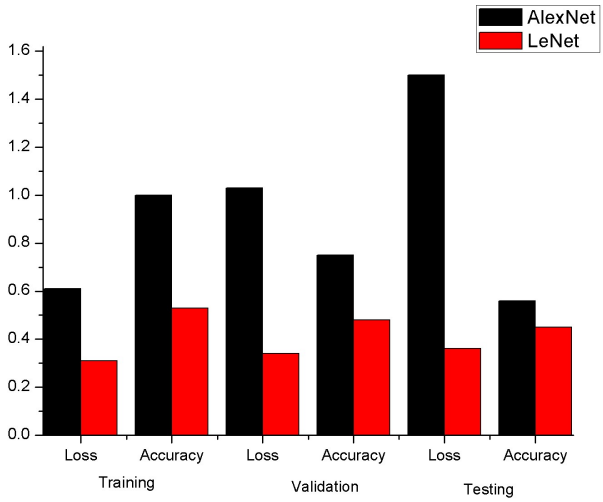
important for the diagnosis of the tumour as a whole. Due to less data availability, data augmentation using cropping is done in this case. The data are divided into training, validation, and testing phases. 80% of the data are for the training phase, the rest is equally divided into the validation and testing phase. Using AlexNet and LeNet the result is as shown in Table 3.

Using the output obtained in Table 3, a graph is plotted between the two methods as shown in Figure 8. It is observed that the accuracy of AlexNet is better than that of LeNet and loss pattern between both methods is the same. Though AlexNet consumes more time than LeNet it can be seen that accuracy is better as the number of layers in AlexNet is more than that of LeNet. It is expected that using a convolutional layer, loss, and accuracy can be better. Due to small datasets even after cropping, these types of methodologies could not perform as expected.

Table 3 Output for classification into metastasis, glioma and meningioma

Method	*Training*		*Validation*		*Testing*		*Computational complexity (approx)*
	Loss	*Accuracy*	*Loss*	*Accuracy*	*Loss*	*Accuracy*	
AlexNet	0.61	1	1.03	0.75	1.5	0.56	39 M
LeNet	0.31	0.53	0.34	0.48	0.36	0.45	2 M

Figure 8 Graph showing accuracy and loss of Alexnet and LeNet for metastasis, glioma and meningioma classification (see online version for colours)



4.6 Classification into HGG and LGG

Using U-Net, the image is classified into HGG and LGG. Images are obtained from BRATS 2017 challenge. Here, we are trying to correctly classify images into HGG and LGG in an unsupervised manner. It is necessary to know the grade or stage of

the tumour as treatment could be more effective. In this paper, the images are first preprocessed using median filtering and followed by U-Net. BRATS data consists of patient data consisting of 155 slices for each patient. The total number of patients for HGG is 210 and LGG is 75. For our experiment, 20 patient data have been taken where a total of 3,100 slices were taken. The images are saved in png format for our purpose.

The output is given in Table 4 and graph is plotted in Figure 9. It is observed that the accuracy of AlexNet is better than that of U-Net. However, loss shows better learning in the case of U-Net. Using CNN can be a stepping stone in image classification. It can be observed that with the increasing number of convolution layers the performance becomes better. Loss can be further improved, and make the network learn better.

Figure 9 Graph showing accuracy and loss of AlexNet and U-Net for HGG and LGG classification (see online version for colours)

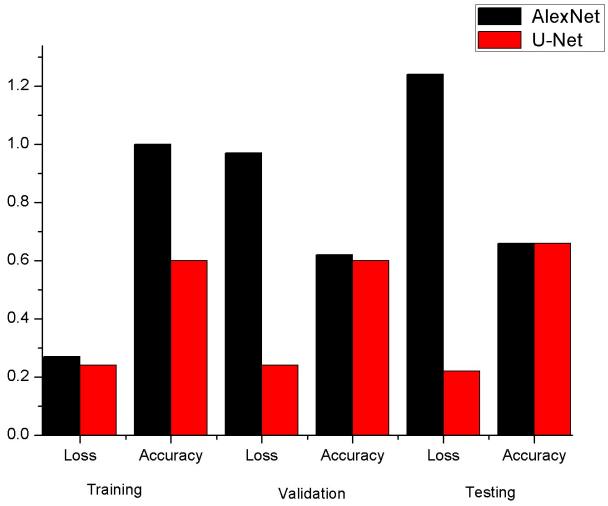


Table 4 Output for classification into HGG and LGG

Method	*Training*		*Validation*		*Testing*		*Computational complexity (approx)*
	Loss	*Accuracy*	*Loss*	*Accuracy*	*Loss*	*Accuracy*	
AlexNet	0.27	1	0.97	0.62	1.24	0.66	39 M
U-Net	0.24	0.60	0.24	0.60	0.22	0.66	7 M

5 Conclusions

Many states of the art deep learning techniques have not been explored much. Despite progress in computer vision (CV), the problem of solving CV is not even close. However, people are trying and several methods using CNN have been adopted and applied to achieve success in real world problems.

In this paper, several deep learning methods have been analysed for achieving better accuracy. Even though it failed to give good results in some cases, it is observed

that using CNNs is very effective in the classification of medical images. In our case, we could see that ResNet, DenseNet, and AlexNet give better results than other methodologies. But such Nets need a large amount of data to train. In actual, these type of Nets can outperform any other methods if data preciseness and genuineness is guaranteed. Though the data augmentation technique is used in the case of classification into metastasis, glioma, and meningioma, the data are still less so could not obtain better accuracy. Therefore, data augmentation like mirroring and cropping can be used in many ways in the future. Works related to classification into type and grade of the tumour can also be explored more to help in the segmentation and diagnosis process.

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